

Physiology lec. 7

Sympathetic & Parasympathetic and Neurotransmitters

Sympathetic & Parasympathetic

A-Sympathetic NS

- Known as fight or flight responses
- Sympathetic is responsible for Emergency cases (Stress & Alarm)
- What Does Sympathetic do during Emergency cases?

Action: **ACVMBBB**

- 1-**A**cceleration of heart rate
- 2-Increase **C**ardiac output
- 3-**V**asoconstriction in blood vessels
(narrowing blood vessels)

4- increase **B**lood glucose, **B**lood pressure,
Blood Coagulation

5-Increase **M**etabolic rate

Parasympathetic Nervous System

-Known as Rest & Digest system

-Does the opposite of Sympathetic Nervous System

From the work of Para and sympathetic nervous systems They lead to Homeostasis (Internal body balance)

NeuroTransmitters

-Substances produced by Nervous system

-Life Cycle of NeuroTransmitters

1-Synthesized

2-Stored

3-Released

4-Bind to receptors to perform actions

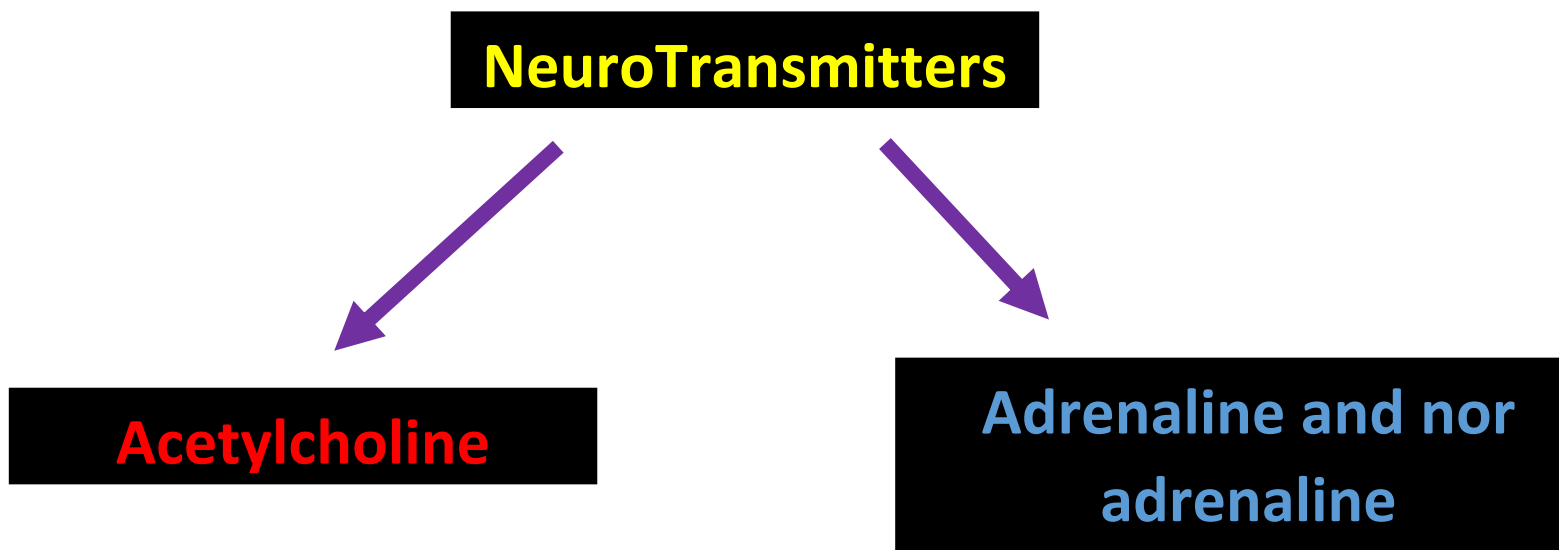
5-Destruction into smaller in units

6-Resynthesized again

-Due to this cycle neurotransmitters doesn't end from the body as they keep getting resynthesized

Example of Neurotransmitters:

**Dopamine, Adrenaline, Nor adrenaline,
Acetylcholine**



Acetylcholine

-**Acetylcholine** found in Synapse / Ganglia (post & pre) / Motor end plate (myoneural junction)

- **Acetylcholine** Normally produced from Parasympathetic Ganglia (post and pre)
But Preganglionic sympathetic nerve also produces acetylcholine

Some Postganglionic sympathetic nerves produce acetyl choline which are :

Sweat gland, piloerector muscles and few blood vessels

(Piloerector muscles are the muscles which move the hair due to goosebumps الرعشة)

-**Acetylcholine** is formed from **Choline** and **acetate** inside **Nerve endings (Axon terminals)** and stored in synaptic vesicles

-**Acetylcholine** is destroyed to Choline and acetate by **cholinesterase enzyme**

-Receptors of Acetylcholine are **cholinergic receptors**

Those receptors are divided into **muscarinic receptors** and **Nicotinic receptors**

A-Muscarinic receptors are found in:

a-Postganglionic nerve of parasympathetic nerves

b-cholinergic neurons of sympathetic sweat glands, piloerector muscles and few blood vessels

B-Nicotinic receptors are found in:

a-Synapses between pre and post ganglionic neurons of sympathetic and parasympathetic

b-Found in membranes of skeletal muscles fibers

NOTE:

Nicotinic receptors are affected by nicotine

Muscarinic receptors are affected by muscarine which is found in toads

What happens if Acetylcholine increase/decrease in Body?

**Increase: will lead to decrease in Heart rate causing Heart failure (Due to Cardiac Block
(فشل في عضلة القلب)**

-Increase in acetylcholine is treated by adrenaline & atropine

Decrease: leads to paralysis due to lack of muscle contraction

Adrenaline and nor adrenaline Neurotransmitters

- They are produced from medulla and sympathetic nervous system
- Produced by medulla to provide organs with no synaptic supply to have their share of adrenaline and nor adrenaline such as endothelium
- They are formed by the help of tyrosine amino acid

Difference Between Adrenaline and nor adrenaline in function

Action/ hormone	Adrenaline	Nor adrenaline
Cardia output	Increase	Slight decrease
Peripheral resistance	Decrease	Increase

Blood glucose	increase	Little effect
Blood pressure		Increase

-Adrenaline differ from nor adrenaline in structure where adrenaline has methyl group while nor adrenaline doesn't have methyl group

-Adrenaline receptors are called Adrenergic receptors which are divided into

Alpha and Beta receptors which has 4 types

Alpha 1, Alpha 2 , Beta 1 and Beta 2

-Adrenaline is destructed by Monoamine oxidase

Adrenaline 2nd name: epinephrine

Nor adrenaline 2nd name: Nor epinephrine

Differences Between Alpha and Beta receptors on organs

1-Alpha receptors

VIPDB

- a-Vasoconstriction
- b-Iris dilation (widening of Pupil in eye)
- c-Pilomotor muscles contractions
- d-Decrease motility and secretion of intestine
- e-Bladder sphincter contraction

2-Beta receptors

BIGVCLUB :

- a-Bladder relaxation
- b-Intestinal relaxation
- c-Glycogenolysis (breaking down glycogen)
- d-Vasodilation by beta 2

e-**C**ardioacceleration by beta 1

f-**L**ipolysis (breaking down of lipids)

g-**U**terus Relaxation

h-**B**ronchodilation (widening air ways which are bronchi and bronchioles)